Brain and Cognition 80 (2012) 45–52

Contents lists available at SciVerse ScienceDirect

Brain and Cognition

journal homepage: www.elsevier.com/locate/b&c

Surface EEG shows that functional segregation via phase coupling contributes to the neural substrate of mental calculations

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ARTICLE INFO

Article history:
Accepted 5 April 2012
Available online 23 May 2012

Keywords:
Mathematical cognition
EEG
Phase synchrony
Graph theoretic analysis

ABSTRACT

Multichannel EEG traces from healthy subjects are used to investigate the brain's self-organisation tendencies during two different mental arithmetic tasks. By making a comparison with a control-state in the form of a classification problem, we can detect and quantify the changes in coordinated brain activity in terms of functional connectivity. The interactions are quantified at the level of EEG sensors through descriptors that differ over the nature of functional dependencies sought (linear vs. nonlinear) and over the specific form of the measures employed (amplitude/phase covariation). Functional connectivity graphs (FCGs) are analysed with a novel clustering algorithm, and the resulting segregations enter an appropriate discriminant function.

The magnitude of the contrast function depends on the frequency-band (θ, α_1, α_2, β and γ) and the neural synchrony descriptor. We first show that the maximal-contrast corresponds to a phase coupling descriptor and then identify the corresponding spatial patterns that represent best the task-induced changes for each frequency band.

The principal finding of this study is that, during mental calculations, phase synchrony plays a crucial role in the segregation into distinct functional domains, and this segregation is the most prominent feature of the brain's self-organisation as this is reflected in sensor space.

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1. Introduction

The brain's remarkable ability to self-organise, not only in response to sensory stimulation but also as a result of endogenous information processing, has repeatedly been demonstrated in the past. Modern neuroimaging techniques facilitate the observation of pattern formation in intact brain and at multiple time and space scales. Recent methodological innovations have opened the possibility of studying and tracking emerging patterns that are based on various signals and derived representations (Mitra & Bokil, 2007). In this way, various cognitive tasks have been decomposed into different sub-processes, with the underlying neural mechanisms characterised in sufficient detail. The prevailing notion of a dynamical complex system, whose behaviour is governed via abrupt phase transitions, has led to the systematisation of coordination principles (Kelso, 1995). The interplay of functional segregation into local brain areas and global integration through distant interactions is considered the hallmark of self-organisation (Srinivasan, Russell, Edelman, & Tononi, 1999).

Mathematical thinking as a cognitive process activates not only local but also spatially distributed cortical networks, depending upon task specificity and complexity (Micheloyannis, Sakkalis, Vourkas, Stam, & Simos, 2005). Mathematical thinking is also strongly correlated with working memory capacity, language function and general intelligence. Simpler arithmetic tasks, such as simple addition, simple subtraction and multiplication, are correlated with language and long-term explicit semantic memory (Dehaene, Molko, Cohen, & Wilson, 2004). Comparisons between quantities primarily activate the bilateral intraparietal sulcus (Delazer et al., 2003; Barth et al., 2006). Frontal and parietal regions are the most common regions to be reported in the literature to be related to mathematical thinking (Barnea-Goraly, Eliez, Menon, Bammer, & Reiss, 2005; Dehaene, Piazza, Pinel, & Cohen, 2003; Rivera, Reiss, Eckert, & Menon, 2005). In addition, an asymmetry between the left and right hemispheres, with a predominance of the left hemisphere in simple arithmetic calculations, has been reported (Rivera et al., 2005; Zhang, Zhang, Zhang, & Li,

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http://dx.doi.org/10.1016/j.bandc.2012.04.001

2005). Two-digit multiplications are rather difficult calculations that demand retrieval of simple mathematical facts and, in addition, require adherence to a series of mental calculation procedures that exemplify higher order mathematical thinking (Gruber, Keil, & Muller, 2001). These processes are expected to engage a network of areas that are responsible, among others, for language functions, including long-term (consolidated) and verbal memory (Dehaene, Spelke, Pinel, Stanescu, & Tsivkin, 1999; Delazer et al., 2003) as well as verbal working memory (Micheloyannis et al., 2009).

Functional MRI (fMRI) has been used to precisely localise the non-verbal quantity representation system. The results indicate that numerical tasks typically involve a distributed network, including frontal cortex areas and left/right parietal lobes (Chochon, Cohen, Van De Moortele, & Dehaene, 1999; Dehaene et al., 1999; Pesenti, Thioux, Seron, & De Volder, 2000; Piazza, Giacomini, Le Bihan, & Dehaene, 2003; Pinel, Dehaene, Riviere, & Lebihan, 2001; Pinel, Piazza, Le Bihan, & Dehaene, 2004; Rickard et al., 2000). Hemodynamic brain imaging techniques, however, lack the temporal resolution necessary to reveal rapidly changing patterns of local neurophysiological responses and functional connections between brain areas that are part of the brain mechanism that support the performance of a specific cognitive task. Aiming at finer temporal scales, EEG studies have started to appear. Changes in local and coordinated activity were investigated using linear and non-linear measures of the EEG signal in different arithmetic tasks (Micheloyannis et al., 2005). The results indicated that the pattern of neural synchrony varies with the complexity of the performed task, depending on the frequency band, and corresponds to a distributed set of neural networks. Event related potential (ERP) paradigms have also been adapted for the exploration of brain activity during mental arithmetics (Duhong, Jian, & Hongtao, 2011; Wang et al., 2007).

Having in mind to study math-related cognition as a brain state manifestation, we adopted an experimental design that deviates from the traditional approach in which a particular simple arithmetic operation is targeted at. We incorporated more demanding tasks, namely the multiplication of two-digits numbers and the four-digits comparison, instead. These two mathematical tasks depend upon neurophysiological processes that are known to differ regarding their nature and the distribution of the activated brain regions. According to several neuroimaging studies, comparison is a more localised procedure (activating intraparietal brain region) while difficult (two-digits) multiplication demands the activation of a widely distributed network (Micheloyannis et al., 2005 ; Dehaene, 1996; Pinel et al., 2001). The initial motivation was to describe the functional (re)organisation, during the execution of these two tasks, by detecting neural networks according to a graph theoretic perspective (Bullmore & Bassett, 2011).

The principal scope of this study was to investigate whether the emergence of functional clusters is an important mode of pattern formation underlying the cognitive load of two mental calculation tasks and whether these clusters have sufficient signal deviation from the baseline condition. Specifically, we considered the multichannel EEG recordings during rest and active conditions (subjects performing either difficult multiplication or number comparisons) as observations from two different states of the (same) brain, which could be discriminated by a common classification algorithm, and searched for a specific state-representation that would facilitate the best discrimination. Discriminability was measured for different types of coupling (linear vs. nonlinear) and different forms of covariation (amplitude/phase). Because discrimination was formulated independently of a given state representation, it could be expressed in a common scale and accompanied by its statistical significance (calculated using the distribution of random state-assignments). The resting-state vs. active-state classifications with the most prominent separation were identified independently for each frequency-band, and prototypical topographies (representative segregations) best describing each state were extracted and characterised.

This comparative study appointed phase-synchrony as the most important aspect of cognitive processes that underlie mental calculations and showed that segregation into distinct functional networks was an important dimension of task-induced changes. It must, however, be stressed that these interrelationships refer to nodes representing EEG sensor signals that may only approximately relate to actual neural generators.

This paper is organised as follows. In Section 2, after a short description of the experimental data, we describe our procedure for validating class-separability, and we briefly describe the ways to test task-induced differences. Section 3 is devoted to the results from the comparative study of neural synchrony. Section 4 discusses the importance of our findings and their correlation with recent relevant literature, and Section 5 includes the conclusions.

2. Methods

2.1. The experimental data

2.1.1. Subjects

The present study concerned 18 right-handed volunteers (aged: 21–26 years, mean 23 years, who were Medical School students in the University of Crete) during the performance of arithmetic tasks differing in the nature and level of complexity. All of the participants signed an informed consent form after the procedures were explained to them, had normal or corrected-to-normal vision and reported no history of verbal or non-verbal learning disability (all medical students in Greece go through a rigorous national testing process that includes advanced math and written composition).

2.1.2. EEG recordings

EEG was recorded continuously while subjects were performing one of the two arithmetic tasks: (1) four-digit number comparisons (e.g., 5467 vs. 6689; numbers in "different pairs" differed by less than 20%) in which the position of the larger number varied randomly, and participants had to raise their left index finger if the number on top was greater and had to raise their right left index finger otherwise; (2) two-digit multiplication (e.g., 34×23, 49×32), in which participants were allowed 10 s from the time the number stimuli were presented on the screen until either a fixation spot was presented (in the number comparison task) or a number was briefly flashed on the screen (in the multiplication task), and they had to respond as to whether the number coincided with the correct result or not. An additional EEG baseline was recorded during a passive viewing condition (i.e., participants simply fixated at the centre of the computer screen on a small star). Stimuli were presented on an LCD screen located in front of the participants at a distance of approximately 80 cm, subtending 2–3° and 2–4° of the horizontal and vertical visual angle, respectively. Each stimulus was always present for 1.5 s, while a fixation star was presented during the ISI for a 0.5 s duration. The three tasks (control, multiplication and comparisons) were delivered in trials of random order to the subject, who then had 10 s to execute each task using the numbers presented on the screen (or simply to rest in the case of the first task). During the comparison task we were presenting, successively, different pairs of 4-digits numbers, since each single comparison was accomplished in a short time. The subject indicated its response via finger rising, which was recorded via an optical switch, and the visual stimulus was switched to the new comparison. Fig. 1 provides a schematic description of the experimental flow. An experienced neurophysiologist (SM) was inspecting the signals on the monitor of the recording PC and decided

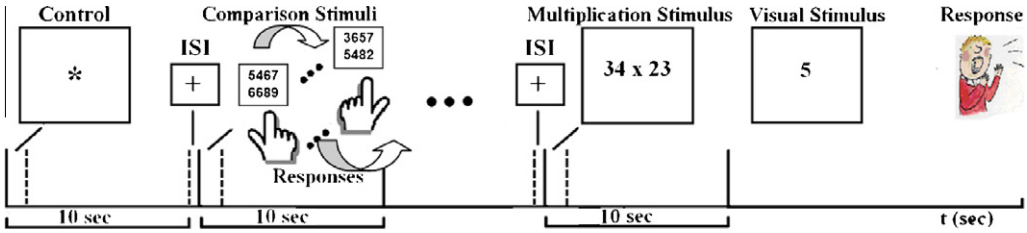

Fig. 1. Schematic illustration of the experimental time course. The arrows indicate that different pairs of 4-digits numbers were successively presented to the subject.

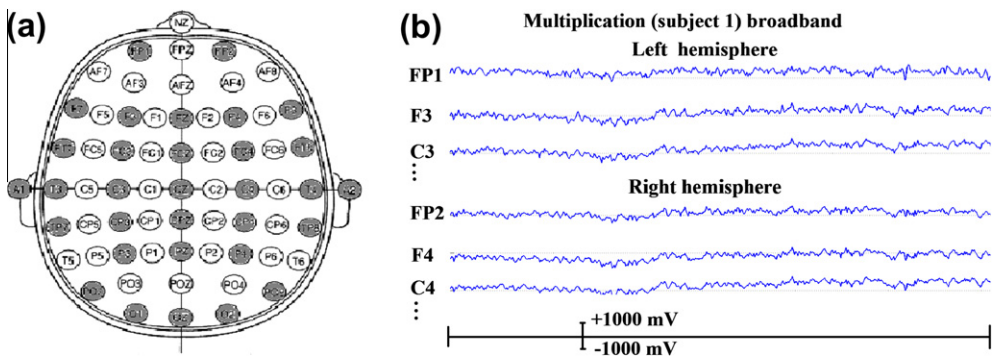

Fig. 2. (a) Scalp topography of EEG-recoding sites and (b) A 1 s (512 samples) segment of an unfiltered multichannel signal from subject-1 during the multiplication task.

whether to reject or accept the trials. Trials contaminated by artefacts (e.g., blinks) were excluded from further analysis. We should underline here that only- trials associated with correct responses were analysed, since wrong answers could be associated with manifold reasons (e.g. loss of concentration) some of which were totally unrelated to the dynamical (sub)processes of math calculation we would like to explore. The performance of subjects, in both tasks, was very high (=100%; as it was expected considering that the participants were highly educated students).

The EEG was recorded from the following 30 electrodes according to the international 10/20 system: FP2, F4, FC4, C4, CP4, P4, O2, F8, FT8, T4, TP8, PO8, Fz, FCz, Cz, CPz, Pz, Oz, FP1, F3, FC3, C3, CP3, P3, O1, F7, FT7, T3, TP7, PO7, and A1 + A2 as a reference (see Fig. 2a). Vertical and horizontal eye movements and blinks were monitored through a bipolar montage from the supraorbital ridge and the lateral canthus. A connected ear-lobe electrode (A1 + A2) was used as reference electrode. The A1 and A2 electrodes were positioned on ear-lobes, far from the scalp electrodes (and far from muscles). The signals were amplified using a set of Contact Precision Instrument amplifiers, filtered online with a (0.1–200) Hz band pass and digitised at 500 Hz. Artefact-free epochs of 8 s were selected via visual inspection for each subject and for each recording condition (for active tasks, data within the time window corresponding to early evoked response were avoided). By applying *ICA* (Delorme & Makeig, 2004), we eliminated artifactual activity (i.e. components associated with eyes, heart or muscles) and transient response. Using a zero phase pass-band filter, the multichannel EEG-signal was split into 5 different bands, which are traditionally defined and denoted as follows: $\boldsymbol{\theta}$ (4–8 Hz), $\mathbf{a}_1$ (8–10 Hz), $\mathbf{a}_2$ (10–13 Hz), $\boldsymbol{\beta}$ (13–30 Hz), and $\boldsymbol{\gamma}$ (30–45 Hz).

2.2. Exploratory analysis

2.2.1. Contrast function

Following the popular pattern-analytic convention (Duda, Hart, & Stork, 2000), two different sets of patterns/objects $\{X_i\}$ and $\{Y_j\}$ can be compared in three steps. We first established an appropriate pairwise dissimilarity measure $D(X_i, Y_i)$. This measure is then applied to all possible pairs to compute the inter-set scatter ($\mathrm{IS}_{\mathrm{control}\leftrightarrow\mathrm{active}}$) and the two within-set scatters ($\mathrm{WS}_{\mathrm{control}}$ and $\mathrm{WS}_{\mathrm{active}}$). The computed quantities are finally combined to express the set-difference, as follows:

$$J = J(\{X_i\},\{Y_j\}) = \frac{\mathrm{IS}_{\mathrm{control}\leftrightarrow\mathrm{active}}}{\mathrm{WS}_{\mathrm{control}} + \mathrm{WS}_{\mathrm{active}}} = \frac{1}{2}\frac{\sum_{i=1}^{N}\sum_{j=1}^{N} D(X_i, Y_j)}{\sum_{i=1}^{N}\sum_{j>i}^{N} D(X_i, X_j) + \sum_{i=1}^{N}\sum_{j>i}^{N} D(Y_i, Y_j)} \quad (1)$$

As implied by the previous equation, the multichannel EEG-traces from the control condition constitute the first set of objects $\{X_i\}$, while the traces recorded during one of the mathematical-tasks populate the second set $\{Y_j\}$. The contrast-function measures the class separability, with the *J*-index taking the value of 1 when classes are highly overlapped and growing monotonically as classes are getting more distinguishable. The given formulation is as generic as possible, incorporating the "feature extraction" step within the computation of the dissimilarity-measure.

The statistical significance of a specific value for the *J*-index can be calculated based on a randomisation procedure (Good, 2000; Goulden et al., 2010), which proceeds by first splitting the objects (i.e., all EEG-traces from rest and active conditions) at random into two groups and repeating the computations for *J* multiple times to form a baseline distribution for the *J*-index indicative of random partitioning. Finally, the original value of the *J*-index is compared to the derived baseline distribution, and this comparison is expressed via a *P*-value.

After selecting an appropriate representation for a brain state and establishing the corresponding dissimilarity, the $N = 18$ patterns of a mental calculation task can be compared against the patterns of the baseline condition. In this study, the brain-state representation is a grouping list (which denotes a specific clustering of electrodes), and the Variational Information metric (VI) is the appropriate dissimilarity measure. The former reflects a specific realisation of the brain's functional segregation, while the latter facilitates the pairwise comparison of distinct segregations. If we neatly denote, the brain-state representation (derived via the

algorithmic steps provided in Appendix B) as a mapping that $X_i \rightarrow \mathbf{c_i}$ and $Y_j \rightarrow \mathbf{c'_j}$, the above equation reads as follows:

$$J = J(\{X_i\},\{Y_j\}) = \frac{1}{2}\frac{\sum_{i=1}^{N}\sum_{j=1}^{N} VI(\mathbf{c_i},\mathbf{c'_j})}{\sum_{i=1}^{N}\sum_{j>i}^{N} VI(\mathbf{c_i},\mathbf{c'_j}) + \sum_{i=1}^{N}\sum_{j>i}^{N} VI(\mathbf{c_i},\mathbf{c'_j})} \quad (2)$$

Next, the contrast-function quantifies differences in functional segregation between the rest and the active condition (a schematic illustration of the computations involved can be seen in Supplementary Fig. S2). We should note here that the passage from EEG signals $\{X_i\}$ to clusterings $\{\mathbf{c_i}\}$ includes the intermediate step of estimating the functional connectivity patterns. This step is described below.

2.2.2. Functional connectivity patterns and their graph representations

To accommodate the various aspects of neural synchrony, we employed different functional connectivity estimators. Specifically, three representative algorithms for estimating *Coherence*, *Nonlinear Interdependence*, and *Phase Locking Index* were employed. Every estimator takes as input a pair of time series recorded at distinct sites and derives an estimate of the strength for the corresponding functional interaction. Such an interaction can have either linear or nonlinear characteristics and can take the form of either amplitude or phase covariation. The estimators were selected to span a wide range of functional connectivity, with the ultimate goal (by comparison) to provide a deeper understanding of the interregional coordination during a mental calculation. Specifically, *coherence* (COH) is a linear measure working in the frequency domain and incorporating both amplitude and phase characteristics. On the contrary, Quiroga's *Nonlinear Interdependence* (NI) is a nonlinear measure based on similarities between trajectories of the underlying dynamical systems (Quiroga, Kraskov, Kreuz, & Grassberger, 2002; a new publicly available MATLAB toolbox for related nonlinear analysis is provided in Beste, Otto, & Hoffmann (2010); a review on dynamic connectivity in neural systems is provided in Breakspear (2010). Finally, the *Phase Locking ind*ex (PLI) neutralises the effects of amplitude variations, utilises Morlet's wavelet transform to estimate instantaneous phases and attempts a statistical aggregation of phase-differences (see Appendix A). Each estimator was applied (using signals filtered within a specific frequency band) to every possible pair of electrodes. The derived measurements have a natural graph-theoretic representation via a fully connected weighted graph, called the functional connectivity graph (FCG), where nodes correspond to electrodes and links are weighted according to the strength of the functional coupling.

Because functional segregation is among the brain's most prominent self-organisation characteristics (Tononi & Edelman, 1998), we attempted to explore its putative role in mental calculations using a brain state representation that would emphasise this possibility. Each FCG graph was first entered into a graph-segmentation algorithm (as recently adopted for an EEG-synchrony study (Dimitriadis, Laskaris, Del Rio-Portilla, & Koudoumis, 2009)) and functionally coupled sites were grouped into mutually exclusive clusters. The induced grouping was then treated as the representation that could be used in our contrast function. Having all of the clusterings available, we measured the contrast of the control condition against (each one of) the cognitive tasks, using Eq. (2).

2.2.3. Prototypical functional segregations

Representative patterns of functional segregation best reflecting the three brain states were derived based on all of the available clusterings and then were presented topographically in a contrasted manner to facilitate comparisons (see Fig. 3). For this derivation, we exploited the fact that VI is a true metric in the space of clusterings and incorporated it in a *vector-median* estimator that identifies the "centroid" of a given set of clusterings (Astola, Haavisto, & Neuvo, 1990). For each one of the three conditions, the ensemble of $N = 18$ clusterings $S = \{\mathbf{c_1}, \mathbf{c_2}, \ldots, \mathbf{c_N}\}$ was used in the following computation of prototypical segregation $\mathbf{c_m}$:

$$\mathbf{c_m} = \arg\min_{\mathbf{c_j}\in S} \sum_{i=1}^{N} VI(\mathbf{c_j},\mathbf{c_i}) \quad (3)$$

The above estimator bears some similarities with the notion of *median-graph*, which is the graph that has the minimum average distance to all graphs in a given set (Bunke, Gunter, & Jiang, 2001).

3. Results

3.1. Functional connectivity graph (FCG) representation

Three different versions of an FCG were utilised as alternative representations (FCG^{COH}, FCG^{NI}, FCG^{PLI}), depending on the employed functional connectivity estimator. For each type of synchrony tested, the necessary parameters (for example, the *embedding-dimension* and *time-lag* in the case of NI) were optimised independently for each frequency-band to achieve the maximal task-discrimination score for the J function (Laskaris, Zafeiriou, & Garefa, 2009). Supplementary Fig. S1 demonstrates this type of exhaustive search for the cases of the PLI-estimator (which depends solely on the parameter δ_j (see Appendix A)) and the NI-index.

By clustering the nodes of each single FCG, patterns of functional segregation were derived for every frequency-band under examination. These groupings were then used to estimate the contrast between a control and an active condition. Table 1 includes the corresponding *J*-values, which express task-induced differences in functional segregation. The elevated *J*-scores ($J > 1$) clearly indicate that self-organisation through group formation is an important tendency in our data. Interestingly, this tendency is better portrayed through the connectivity measurements derived with the phase-locking estimator. In all frequency bands, phase synchrony appears to be the neural underpinning of functional segregation in both cognitive tasks. Statistically significant changes did appear via the NI-estimator, but not with the COH estimator (except in the gamma band). This experimental fact points to the nonlinear-character of the interactions that are involved.

3.2. Functional-segregation prototypes

Because the node-clustering of PLI-based FCGs resulted in the most promising representation (high *J*, while $P < 0.001$) for all of the frequency-bands, we included in Fig. 3 the corresponding partitions, which portray the three tasks best (see Section 2.2.3). The derived prototypical-segregations for the resting state and for both mathematical tasks are listed in the different rows of Fig. 3. In each individual topography, the groups formed are indicated using different colours. The colour code denotes the ranks of the groups, with red associated with the most compact group (and hence the strongest functional cluster). In the inset associated with each topography, the values of the Cohesion-index (an index expressing cluster compactness, as described in Appendix B) are listed for the groups.

The most important trend is that during math calculations, coherent functional groups are emerging. Most of these groups are well localised (i.e., they correspond to nearby brain regions) with a Cohesion-index higher than the groupings that correspond to the control-state. The only exception in the previous observation is the fourth group (coloured light-blue), which is formed in the α_1 and α_2 bands during the comparison task. Interestingly, during the control condition, the whole brain behaves as a single functional identity (or, in a few cases, a second, weakly-connected component

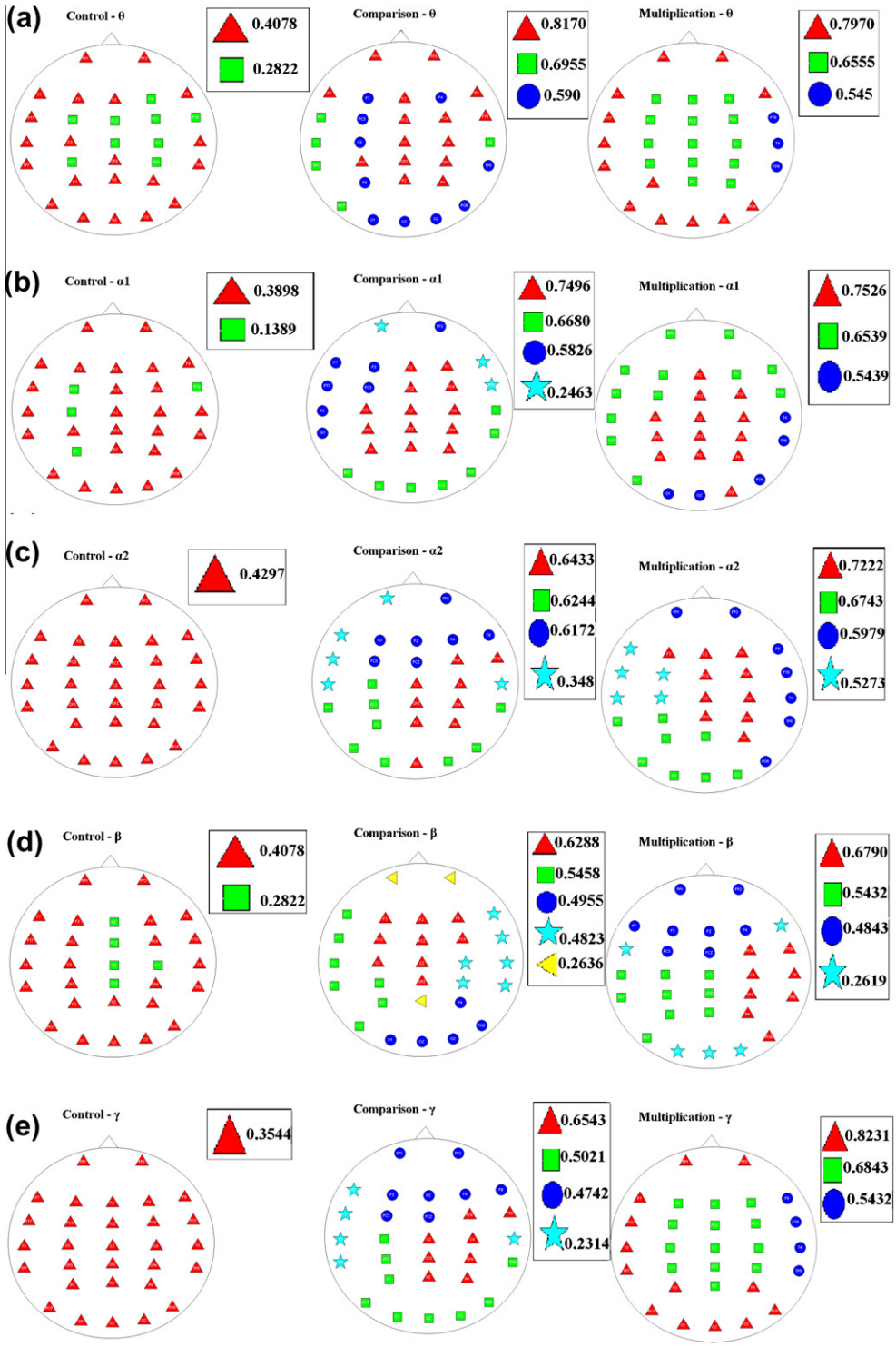

Fig. 3. Prototypical functional segregations extracted with the vector-median estimator. Each row corresponds to a different frequency band, while each inset lists the degree of cohesion for all of the emerged clusters. A dual colour-shape code has been adopted to denote the relative ordering of derived clusters according to the Cohesion index (inset list). In each node-clustering, the red triangles correspond to the most important functional network, the green rectangles to the second most coherent network, etc. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

appears). The above trends were observed consistently across subjects, with only subtle differences among the individual clusterings. More detailed observations can be made by considering the strength of clustering tendencies (Cohesion-level). Focusing, for example, on the α_1 band, we can observe two clusters for the control condition, with Cohesion-indexes 0.3898 and 0.1389, and four clusters during the comparison task, with Cohesion-indexes ranging from 0.2463 to 0.7496. The values included indicate that the "light-blue" cluster, formed within the α_1 band during the comparison task, is less compact than the first cluster extracted during the control condition. Moreover, the Cohesion-index of the three clusters formed within the α_1 band during the multiplication task

Table 1
Contrast-function J for the different functional segregation patterns.

	Control-comparison			Control-multiplication		
	COH	NI	PLI	COH	NI	PLI
θ	2.9067	3.0512**	3.7894**	3.0455	3.1567*	3.7574**
α_1	2.7231	2.8345*	3.8831**	2.8534	3.1045**	3.8211**
α_2	2.9932	3.0534**	4.2484**	2.9245	2.9234*	4.2578**
β	2.9267	3.1023*	4.4427**	3.1023	3.1789**	4.2773**
γ	3.0167*	3.1345**	4.2085**	3.054*	3.1034**	3.7754**

** $P < 0.001$.
* $P < 0.01$.

ranges from 0.5439 to 0.7526. This result means that all of the clusters are more coherent than the "best" cluster detected in the α_1-band under the control condition.

In summary, node-clustering of PLI-based FCGs provided vivid evidence about functional segregation in the brain during both math tasks.

4. Discussion

We studied pattern formation associated with mental calculations based on derivations that rely on the raw EEG signal. Being aware of the convoluted nature of the signal at the sensor level, we searched for changes (associated with math calculations) in the patterning of the neural synchrony and aimed at a topographic description and characterisation of the underlying phenomena.

Before discussing our results, we first set the methodological foundation of our work as follows. Our framework is suitable by design for handling pattern-analytic problems via dissimilarity measures and hence encapsulates recent theoretical advances in the pattern-recognition field (Pekalska & Duin, 2005). This study revolves around the core idea of first establishing a single unifying contrast function that could discriminate between a control task and either of two mental tasks (comparison or multiplication) and then searching for the specific descriptor that would lead to the maximal differentiation. Recently, a similar study has appeared comparing various synchrony measures but with a different approach and in a different context (Alzheimer's disease) (Dauwels, Viallate, Musha, & Cichocki, 2009).

First, we established the contrast function (see Eqs. (1) and (2)), which captured the brain's self-organisation tendencies during mental tasks and could be used in many ways. Using J-index (and the accompanying randomisation test), we explored task-induced phenomena in the emerging connectivity patterns. The application of a node-clustering algorithm in these connectivity graphs offered an important descriptor that emphasised the task-induced functional segregations. J-values were dependent on the frequency band (θ, α_1, α_2, β and γ) and the nature of selected connectivity descriptors (Coherence, NI and PLI).

When comparing the prototypical segregations (Fig. 3), the most striking observation is that the resting state is primarily characterised via a single global functional cluster, while in both tasks, functional clusters are formed corresponding not only to well-localised but also to widely distributed networks. The above trends were consistently observed across subjects.

Then, we need to stress that the PLI descriptor contributed to the maximal discrimination between the mental tasks and the baseline condition. Statistically significant changes appeared via the NI-estimator as well, with lower values for the contrast function, but significant changes did not appear with the COH estimator (except in the gamma band). This study is one of the very first times that a comparative study has been attempted in such a cognitive task. The outcome underlines the nonlinear character of the interactions involved during mental mathematics. When seen from the perspective of complex systems, our experimental findings about the role of phase synchronisation can be interpreted as an indication of rather weak coupling strength (Graben, Zhou, Thiel, & Kurth, 2008).

In conclusion, we scrutinised both functional segregation and integration, but always worked in a comparative manner (searching for differences between a mental task and the rest condition). We first showed that phase-coupling best describes the functional interactions that lead to segregation during the two arithmetic tasks. Moreover, a widespread long-range phase synchrony pattern was also found to accompany these mental mechanisms, to subserve integration. Overall, the findings provide additional support to Varela's hypothesis (Varela, Lachaux, Rodriguez, & Martinerie, 2001), according to which the most plausible mechanisms for neural integration are the formation of dynamical links that are reflected as phase-synchronisation in the EEG signals.

5. Conclusions

This study showed that self-organisation through group formation is an important tendency in two math tasks. Interestingly, self-organisation during math tasks is better portrayed through the connectivity measurements derived with the phase-locking estimator. In all frequency bands, phase synchrony appears to be the neural underpinning of functional segregation in both cognitive tasks. The most important trend is that, during math calculations, coherent functional groups are emerging, while during the control condition, the whole brain behaves as a single functional identity.

An obvious follow-up of the present study would be to compare the patterns of task-induced segregations across different age groups (Micheloyannis et al., 2009) and between healthy individuals and patients with mathematical disturbances such as acalculia and dyscalculia. Exploratory analysis focusing on both time–frequency domains (Dimitriadis et al., 2010) and a variety of imaging modalities (including structural MRI, diffusion MRI, functional MRI, and EEG/MEG) can further enhance current knowledge of the neural mechanisms of numerical calculations and can potentially contribute to the understanding of math inefficiency.

Appendix A. Phase Locking Index

PLI computation is based on estimates of instantaneous phase obtained from the convolution of *Morlet* wavelets with the EEG signals $x_i(n)$ filtered within one of the frequency bands under study. The resulting *Dyadic Wavelet Transform* of a discrete sequence $x(n)$ sampled with time spacing δ_t and consisting of N data points ($n = 0, 1, \ldots N-1$) is denoted as:

$$W^X(n,s) = \sqrt{\delta_t/s}\sum_{n'=0}^{N-1} x(n')\psi_{o^*}((n'-n)\delta_t/s) \tag{A.1}$$

where '*' denotes complex conjugation with the consecutive scaled and translated versions of the principal wavelet function, the complex Morlet wavelet $\psi_o(n)$:

$$\psi_0(n) = \pi^{-1/4} e^{i\omega_0 n} e^{-n^2/2} \tag{A.2}$$

where ω_0 is the nondimensional frequency, here taken to be 6 (Torrence & Compo, 1998). A set of different scales s is implied in Eq. (A.1). Writing the scales as fractional powers of two yields the following:

$$s_j = s_o 2^{j\delta_j}, j = 0, 1, \ldots, J \tag{A.3}$$

$$J = \delta_j^{-1}\log_2(N\delta_t/s_o)$$

where s_o is the smallest resolvable scale and J the largest scale; our analysis starts by estimating the optimal δ_j for each band/condition (see Section 3.1 and Supplementary Fig. S1)

The instantaneous phase $\phi^{Xi}(n, f)$ is then calculated as follows:

$$\phi^{Xi}(n,s) = \arctan\frac{\mathrm{imag}(W^{Xi}(n,s))}{\mathrm{real}(W^{Xi}(n,s))} \tag{A.4}$$

The successive phase values (originally ranging in $[-\pi\ \pi]$) underwent an unwarping transform to get rid off discontinuities (in accordance with the phase correction algorithm described in (Freeman and Rogers, 2002)). Finally, the PLI index for a pair of signals $x_k(n)$ and $x_l(n)$ recorded at different sites was computed by averaging the instantaneous phase differences

$$\mathrm{PLI}(x_k(f_n), x_l(f_n)) = \left|\frac{1}{N\cdot\Delta s}\sum_{n=1}^{N}\sum_{s=s_1}^{s_2}\ \exp(i(\varphi^{x_k}(s,f_n) - \varphi^{x_l}(s,f^n))\right| \tag{A.5}$$

where N is the number of time points (samples), s_1/s_2 refers to the scale limits, Δ denotes the corresponding range and f the studying frequency band (Lachaux et al., 2000). A zero PLI-value means that the phases of two encountered signals were not coupled at all, while the PLI-value of one corresponds to phase values that are completely synchronised.

Appendix B. Graph-theoretic clustering and VI metric

The summary included here has been adapted from a recent EEG-study (Dimitriadis et al., 2009), where more details can be found. Because a symmetrical FCG was required by the clustering methodology of (Pavan & Pellilo, 2007), in the case of a non-symmetric NI index (Quiroga et al., 2002), the maximum coupling was kept for every pairwise interaction regardless of direction. Coherence and PLI are symmetric by their nature. The calculations of the NI index were performed using the related Matlab-code (synchrom; Quiroga et al., 2002) with the number of nearest neighbours set to 10, the Theiler-correction set to 50 (Theiler 1986) and the remaining necessary parameters optimised, as described in Section 2.2.2 (Supplementary Fig. S2).

A graph-theoretic algorithm was employed for identifying the most-cohesive group(s) of vertices given the weighted similarity (adjacency) matrix G of an FCG. The algorithm is based on the identification of the dominant set of nodes and, when repeatedly applied, facilitates effective clustering in a sequential mode of pairwise relational data. In our case, the calculation takes the form of deriving the n-dimensional vector $\mathbf{y}$ ($n = 30$) that maximises the following objective function (i.e., the set-compactness).

$$\begin{aligned}&\text{maximise } F(y) = \mathbf{y^T G y}\\ &\text{subject to } \mathbf{y}\in\Delta,\ \Delta^n = \left\{\mathbf{y}\in\mathbf{R^n} : y_i \geqslant 0\forall i \vee \sum_{i=1}^{n} y_i = 1\right\}\end{aligned} \tag{A.6}$$

The insets in Fig. 3 list the degree of *Cohesion* (set-compactness) for each of the successively formed clusters.

Having all of the clusterings available (the output of the graph clustering algorithm was a 30-tuple $\mathbf{c} = [c_1,\ldots,c_{30}]$, $c_i \in Z$ (e.g., $\mathbf{c} = [1\,3\,2\,3\,2\ldots\ 1\,1\,1\,2]$) reflecting the graph-partition), we quantified the contrast of the control condition vs. the cognitive tasks for each frequency band using the VI-metric as the dissimilarity measure (Meila, 2007).

Variation of Information (VI) (Meila, 2007) is a novel information-theoretic criterion, which has been introduced for comparing two different clusterings of the same data set and measures the amount of information that is lost or gained in changing from clustering $\mathbf{c}$ to clustering $\mathbf{c}'$. VI is a non-negative, symmetric metric that is more intuitive compared to other relevant measures and moreover, overcomes previous limitations (for example, it satisfies the triangular inequality). VI is defined by the function:

$$\mathrm{VI}(\mathbf{c},\mathbf{c}') = [H(\mathbf{c}) - I(\mathbf{c},\mathbf{c}')] + [H(\mathbf{c}') - I(\mathbf{c},\mathbf{c}')] \tag{A.7}$$

where $H(\mathbf{c})$ denotes the entropy associated with clustering $\mathbf{c}$ and $I(\mathbf{c}, \mathbf{c}')$ denotes the mutual information between the two clusterings, c and c'.

The Matlab code that implements the graph-clustering algorithm and the VI metric can be downloaded from the author's website http://users.auth.gr/~stdimitr/software.html.

Appendix C. Supplementary material

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.bandc.2012.04.001.

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